

Drivers of lodgepole pine recruitment across a gradient of bark beetle outbreak and wildfire in British Columbia

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ABSTRACT

Seedbanks are essential for forest resilience, and disturbance interactions could potentially modify seedbank availability, subsequent forest regeneration patterns, and successional trajectories. Regional mountain pine beetle outbreaks have altered forest structure and seedbanks in fire prone-landscapes across western North America and could influence forest regeneration. We examined the drivers of lodgepole pine recruitment across a gradient of fire severity from fire that burned through gray-phase outbreak conditions in central interior British Columbia, Canada. We considered ten potential drivers of lodgepole pine recruitment including: fire severity, mountain pine beetle outbreak severity, cone abundance, branch retention, open cones, exposed soil, snag fall, snow accumulation, climate moisture deficit, and frost events, as well as the contribution of unburned fire refugia to lodgepole pine recruitment. In 2016, we inventoried 83 plots that spanned gradients of outbreak severity, burning conditions, and fire severity across three fires that burned through lodgepole pine dominated forests with gray-phase conditions that had moderate to high mortality. We used generalized linear models to evaluate drivers of variability in seedling recruitment. Our analyses affirm that wildfire is a critical mechanism for lodgepole pine seedling recruitment. For plots that burned as crown and surface fire, recruitment densities were two or six times greater compared plots that experienced light surface or were fire refugia, respectively. Recruitment was driven by cone abundance indicating the importance of an available seed source. In fire refugia, recruitment was three times greater during the post-outbreak period compared to before outbreaks, suggesting cone senescence from mortality and/or xeriscent-cueing of cones. The unevenness in recruitment across gradients of outbreak and fire severity contribute to the heterogeneity of stand initiation across these landscapes that may be important for mediation of subsequent outbreaks.

1. Introduction

Landscape-scale disturbances such as mountain pine beetle (*Dendroctonus ponderosae*; hereafter MPB) outbreaks and wildfires are ecological agents that shape composition and structure in the forests across western North America. Insect outbreaks and wildfire are increasing in area affected under contemporary climate conditions (Dale et al., 2001; Gonzalez et al., 2010), and overlapping in time and space to create conditions with narrow windows between disturbances – i.e. short-interval disturbances (Turner, 2010; Buma, 2015). In British Columbia (BC), Canada, the extent of the recent MPB outbreaks in the sub-boreal region, circa 1999-present, has exceeded the extent and severity of mortality conditions in the western United States (US) over recent decades (Raffa et al., 2008) and in the historical record for BC (Axelson et al., 2009; Perrakis et al., 2014). When fire occurs in these forests,

there is concern there could be synergistic interactions between fire and the legacy left by the MPB outbreaks that will influence postfire structural legacies, ecological response, and forest resilience (Holling, 1973; Paine et al., 1998; Simard et al., 2011; Johnstone et al., 2016; Talucci and Krawchuk, 2019).

Life history traits and seedbank characteristics of trees are key mechanisms influencing forest regeneration and succession following broad-scale disturbances. Lodgepole pine (*Pinus contorta* var. *latifolia*) is a widespread species across western North American forests that exhibits cone serotiny, the retention of an aerial seedbank within sealed cones past maturation and awaiting an environmental cue (Lamont et al., 1991). Serotiny is a key mechanism facilitating postfire regeneration (Lotan et al., 1985; Muir and Lotan, 1985; Tinker et al., 1994) and development of early seral forest communities. Seed release from serotinous lodgepole pine tends to be pyriscent (fire cued), but can

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also be xeriscent – i.e. opening cued from warm, dry conditions (Lamont et al., 1991), or cone opening triggered by senescence associated with tree mortality (Teste et al., 2011). Pyriscent cones generally trigger a large pulsed seed release, while xeriscent cones tend to trigger seed release from a few cones that experience direct sunlight (Crossley, 1956; Teste et al., 2011).

Postfire seedling density is largely controlled by prefire serotiny levels in lodgepole pine (Lotan, 1976; Muir and Lotan, 1985; Tinker et al., 1994; Turner et al., 1997; Schoennagel et al., 2003); however, severe MPB outbreaks prior to fire may alter typical serotiny responses. Once snags (standing dead trees) drop their needles and transition to what is referred to as the gray-phase of an outbreak (Hopkins, 1909; Wulder et al., 2006; Simard et al., 2011), cones may experience xeriscent cueing. Even with seed release from xeriscent cueing or live trees with non-serotinous cones, regeneration is slow for these shade-intolerant pines due to resource competition (Astrup et al., 2008; de Ville, 2013). When MPB-killed snags reach the gray-phase, the aerial seedbank is reduced by an estimated 45% from xeriscent cueing, squirrel predation of canopy cones, and relocation of cones to the forest floor through branch breakage – i.e. stand “breakdown” (Teste et al., 2011). If MPB outbreaks modify the typical serotiny response to subsequent fire, there is potential for interruption of the usual cueing and postfire regeneration response, creating compound disturbance effects *sensu* Paine et al. (1998), where one disturbance affects the ecological outcomes of a subsequent event. Further, recruitment in unburned locations, or fire refugia, that only experience MPB outbreaks could augment landscape heterogeneity with low density multi-aged cohorts.

Wildfire is an important mechanism for seed release and recruitment in lodgepole pine dominated forests. However, wildfires preceded by MPB outbreaks may alter fire severity (i.e. loss in biomass; Keeley, 2009) and subsequent recruitment response. In the US Northern Rocky Mountains, postfire lodgepole pine recruitment was attributed to prefire cone serotiny and was considered unaffected by the interaction of MPB outbreaks and wildfire (Harvey et al., 2014a, 2014b). However, subalpine forest experiencing moderate- to high-severity wind-throw events followed by moderate- to high-severity wildfire failed to regenerate because the aerial seedbank was exposed to slower moving surface fire, while the combination of less severe short-interval disturbance conditions resulted in robust regeneration (Buma and Wessman, 2011). Areas with high severity outbreak mortality (80–100%), are vulnerable to stand breakdown – branch breakage and wind-throw (Lewis and Hartley, 2005), which relocates the aerial seedbank to the surface. Cones at the surface are exposed to a slower rate of spread during the initial flaming front, which exposes cones to longer flame residence times that can reduce the seedbank through combustion or over heating (Alexander and Cruz, 2012). Prefire snags are more likely to torch and smolder, resulting in consumption of bark, branches, cones, and development of deep char (Donato et al., 2009; Talucci and Krawchuk, 2019). These outcomes suggest that synergistic interactions are dependent on the lag between events and magnitude of the disturbances. When wildfire burns through stands with high mortality from MPB outbreaks, as seen in BC's central interior lodgepole pine ecosystems, it is not clear whether these stands will be resilient – i.e. demonstrate the early successional recruitment expected from their pyriscent life history strategy – or whether they will be sparse, or even fail to regenerate.

Climate conditions, including winter frost events, snow accumulation, and growing season water stress, are known controls of recruitment. In subalpine environments during colder months, frost events can desiccate seedlings (Stuart et al., 1989; Inouye, 2000), while snow accumulation insulates seedlings and provides moisture, thus enhancing survival rates (Batllori et al., 2009; Renard et al., 2016). During warmer months, longer growing periods and warmer conditions boost recruitment in subalpine environments (Clark et al., 2017). However, extended periods of water stress and/or prolonged drought postfire are suggested to preclude and/or delay regeneration (e.g. Harvey et al.,

2016; Hansen et al., 2018). The relationship between postfire climate and recruitment remains largely understudied in BC's sub-boreal forests (Wong et al., 2004).

Here we examine the drivers of lodgepole pine recruitment in early seral forests in central interior BC across a gradient of fire severity generated from burning in gray-phase outbreak conditions. We had two objectives. First, we determine the drivers of seedling recruitment after short-interval disturbances across a gradient of fire severity by considering the effect of ten potential contributing factors: fire severity, outbreak severity, cone abundance, branch retention, open cones, exposed soil, snag fall, frost events, snow accumulation, and climate moisture deficit. Second, we determine the drivers of seedling establishment in fire refugia within fire perimeters, including establishment periods, outbreak severity, cone abundance, branch retention, and open cones. We expected fire to serve as a critical mechanism for seed release and enhanced recruitment across a gradient of fire severity. However, we hypothesized that factors associated with stand breakdown – outbreak severity, cone abundance, branch retention, open cones, and snag fall would act as mechanism that reduce the seedbank and subsequent post-fire recruitment. Factors associated with resource availability on the forest floor, mainly exposed soil, were thought to be a vital mechanism for successful recruitment. Factors associated with postfire snow accumulation, frost events, and climate moisture deficit were hypothesized to either enhance or constrain recruitment depending on the conditions. Addressing these research gaps may inform future decision-making processes for research and forest ecosystem management.

2. Methods

2.1. Study area

Our study area included three wildfire footprints that burned in 2012, 2013, and 2014 within Tweedsmuir and Entiako provincial parks (Fig. 1). These fires burned through forests with substantial prefire mortality from MPB outbreaks. Lodgepole pine forests experienced severe tree mortality about 10–12 years prior from regional MPB outbreaks, which altered the arrangement of live and dead vegetation/fuels across the landscape. These forests are associated with the Sub-Boreal Pine Spruce and Sub-Boreal Spruce biogeoclimatic zones (Meidinger and Pojar, 1991). For the region, temperatures and precipitation were calculated from the monthly means from the 1981–2010 climate normals data (Abatzoglou et al., 2018): the mean annual maximum temperature is 8.5 °C (range: -3.3–19.8 °C), and the mean nighttime temperature is -2.8 °C (range: -11.9–6.7 °C). Annual precipitation totals 507.6 mm, and mean monthly precipitation is 42.3 mm (range: 22.7–60.8 mm) with snowfall during winters months. The landscape is gently rolling with elevation ranging from 873 m to 1057 m. Forest composition and structure of the region is dictated by moisture gradients and disturbance history (DeLong, 1998). Pure lodgepole pine stands occupy the driest portions of the moisture gradient and persist with estimated fire return intervals of 100–150 years (Parminster, 1992). On moister sites with subalpine fir (*Abies lasiocarpa*) and spruce hybrid (*Picea engelmannii* × *glauca*), seral lodgepole pine persists with longer fire return intervals (Cichowski and Morgan, 2009). The wettest pockets are dominated by spruce, such as riparian zones, that experience long periods without fire (Cichowski and Morgan, 2009).

The study area has no road access, and sampling was limited within 10 km radius hiking and boating distances from three remote cabins in the provincial parks. The parks are managed as wilderness and provided an ideal setting to study MPB-fire interactions without the influence of active forest management (e.g. harvest, suppression). All fires received minimal to no suppression activities (Rob Krause and Mike Pritchard, BC Wildfire Service, *personal communication*). The Entiako Lake fire was lightning ignited on August 3, 2012, and burned until

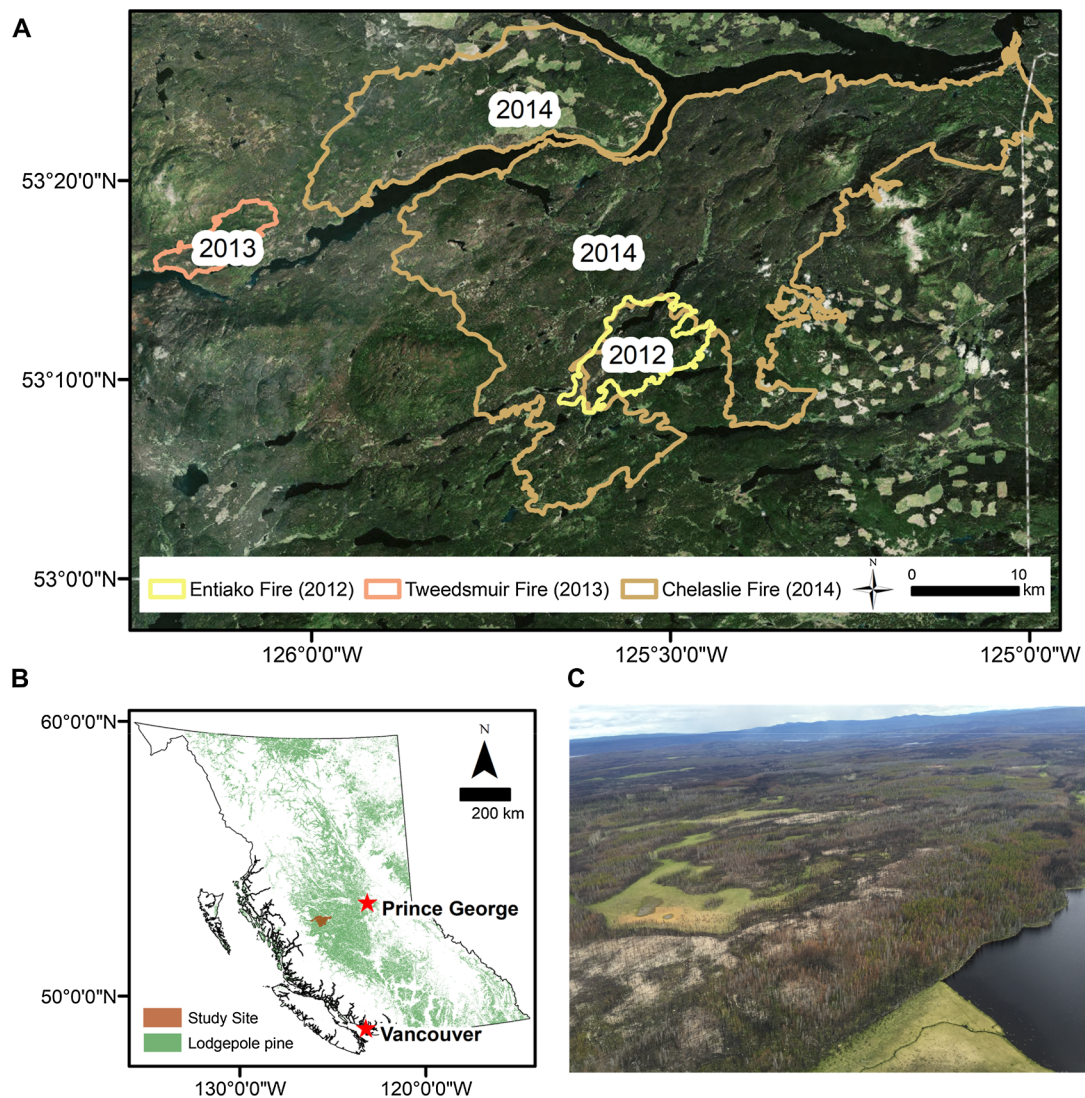


Fig. 1. Geography of the study area. (A) Fire perimeters for three study fires that burned in 2012 (Entiako Fire; 7459 ha), 2013 (Tweedsmuir Fire; 3354 ha), and 2014 (Chelaslie Fire; 133,000 ha). (B) Location of study area and the distribution of lodgepole pine within British Columbia, Canada. (C) Aerial photograph of postfire landscape at Chelaslie River.

September 22, 2012, burning 7,450 ha (BCWS, 2016). The Tweedsmuir fire was lightning ignited on September 9, 2013, and burned until September 16, 2013, burning 3600 ha (BCWS, 2016). The Chelaslie River fire was lightning ignited on July 8, 2014, and burned until October 26, 2014, burning 133,100 ha (BCWS, 2016).

2.2. Sampling design and data

We conducted field sampling from June through August 2016. Data collection captured various stages of development for early seral forest communities from two to four years postfire in areas that were dominated by lodgepole pine (> 75%) in the canopy. Sampling captured a gradient of fire severity and outbreak severity including fire refugia that only experienced MPB outbreak. We conducted an initial *a priori* site selection using digital data that included aerial survey maps of outbreak severity (BCMFLNRO, 2016), burn severity maps based on methods developed by Eidenshink et al. (2007), and vegetation maps based on the Vegetation Resource Inventory, a provincial inventory generated from aerial photos (BCMFLNRO, 2012). Due to the coarse resolution (400 m) of the aerial survey maps of outbreak severity, these served as a rough guideline with the hope of capturing a gradient of outbreak severity (i.e. tree mortality). We used burn severity maps to deliberately

collect data across four classes of burn severity – unburned, low, moderate, and high – that corresponded to a gradient of fire severity, i.e. increasing biomass consumption and loss (Talucci and Krawchuk, 2019) and were split into classes (e.g. unburned, light surface, severe surface, and crown fire) for interpretability. Our generalized fire severity metric is based largely on biomass consumption at the surface and on prefire snags due to the substantial tree mortality that existed prior to fire. Our data collection resulted in replicates of fire severity classes that contained a gradient of underlying MPB outbreak severity. We conducted a final visual assessment on the ground to ensure plots were representative of the surrounding area and lodgepole pine was the primary canopy tree species. Unburned plots, fire refugia, experienced MPB outbreaks but not fire. Fire refugia were located in unburned patches within fire perimeters. We did not observe any biophysical characteristics that suggested these refugia were associated with topographic features or fuel/forest structure variability (i.e., these were likely stochastic fire refugia rather than predictable ones *sensu* Krawchuk et al. (2016)).

In the field, we navigated to pre-selected plot locations and, if they met the *a priori* design criteria, we established plot center for field sampling. UTM coordinates were recorded at the plot center with a Global Positioning System (GPS; Garmin GPSMAP 78 s). Each plot

measured ten-by-ten-meter (100 m^2 ; 0.01 ha). We divided each plot into four quadrants along the north–south and east–west axes identified as NE, SE, SW, and NW. Within each quadrant a one-by-one-meter (1 m^2) subplot was situated at increasing distances from the center of the plot (NE-1 m, SE-2 m, SW-3 m, and NW-4 m) and used to measure seedling recruitment and surface fire effects, described below. We surveyed all canopy trees in each plot, including live trees, snags, and fallen snags (i.e. coarse woody debris) that would have been canopy trees prior to falling. We classified canopy trees as those with a diameter at breast height (DBH) of 7 cm or greater and a height minimum of 4 m. For each canopy tree, we recorded the species, whether it was alive or dead at time of sampling, the DBH to the nearest 0.5 cm, evidence of beetle activity, our metrics associated with stand breakdown, and immediate fire effects as measures of fire severity.

For each canopy tree, we identified the cause of death (i.e., MPB, fire, or other; Appx. Table S1) following protocols outlined by Talucci and Krawchuk (2019) and adapted from Harvey et al. (2013, 2014a). Dead trees identified as MPB-killed had evidence of exit holes and j-shaped galleries specific to MPB. Trees classified as “other-killed” lacked evidence of exit holes and j-shaped galleries specific to MPB, however other evidence indicated prefire mortality including Ips beetle (*Ips pini*) activity, needle loss, sloughing bark, and decay and/or deep char at the base. A tree was classified as fire-killed if it exhibited red needles in the canopy or postfire needle drop and no evidence of prefire MPB or other beetle activity. Most of the prefire tree mortality present was linked to MPB (Appx. Table S1). At the plot-level, outbreak severity was represented as the proportion of prefire-killed trees, which was the combination of MPB-killed plus other-killed trees. We opted to include other-killed trees with MPB-killed, because they were a small portion of the trees sampled and contribute to prefire tree mortality that had the potential to influence stand structure and seedbank locations (Appx. Table S1).

We characterized variables of stand breakdown, e.g., outbreak severity (described above), branch retention, open cones, cone abundance, and snag fall. Branch retention was characterized as the presence or absence of all branch structure, which included fine, moderate, and coarse branches in the canopy. Absence indicated no remaining branch structure on either the tree or the ground in the immediate area of the tree meaning that branches were consumed by fire. Presence denoted at least partial to full branch structure on a tree. Branch retention was aggregated to a plot-level variable as the proportion of trees with branches. Each tree was classified according to whether it had a majority of cones open, scored as one for yes and zero for no, and then calculated as proportion of trees with open cones per plot. We counted cones retained in the tree canopy using binoculars and a tally counter, to serve as a general estimate of potential seed availability. We calculated a plot-level variable of cone abundance as the mean cones per tree. Our snag fall metric accounted for all prefire snag fall, which was considered a component of stand breakdown that could influence seedbank availability. Prefire snag fall was distinguished from postfire snag fall based on presence of charring at the break point of the snag and was absent on postfire snag fall. We calculated a plot-level metric for the proportion of prefire snag fall indicating re-location of seedbank to the surface prior to fire.

We characterized fire severity using nine immediate fire effects metrics including scorch/char height on live/dead trees, percentage of scorch/char on live/dead trees, level of deep char live/dead trees, depth of remaining litter/duff, exposed mineral soil, remaining litter, and terrestrial surface char. Fire behavior produces immediate fire effects that are measured to characterize fire severity. With high rates of prefire mortality, fire severity refers to biomass consumption and loss at the surface and on live/dead trees/snags (Keeley, 2009; Talucci and Krawchuk, 2019). Because plots were surveyed between two to four years postfire, they captured various early seral stages of postfire litter accumulation and vegetative regrowth. We estimated the height of maximum bark scorch/char to the nearest 0.5 m with a four-meter

measuring stick. We visually estimated and recorded the percentage of the tree covered with scorch/char across the height and circumference and assigned a category for the percentage of deep charring on the full height and circumference of the tree (0 percent, less than 50 percent, or ≥ 50 percent). Within each subplot, we quantified and recorded surface fire effects metrics including duff depth, exposed mineral soil, remaining litter, and surface char based on a visual estimate of the percent cover. We measured the combined litter and duff depth to the nearest millimeter with a ruler in two opposing corners of each subplot. The proportion of exposed mineral soil was considered a contributing factor for seedling recruitment in our analyses. In the field, we assigned each plot a fire severity class: unburned, light surface, severe surface, or crown fire based on visual inspection of fire effects and biomass consumption. Each fire effect was aggregated to a plot-level metric. We evaluated our fire severity classes with scatter plots and a classification tree using our plot-level fire effects to confirm their validity and retained those classes for analysis (Appx. Fig. S1).

We represented postfire climate as annual frost-free days, annual snow accumulation, and annual climate moisture deficit, acquired from the Climate BC dataset with a gridded spatial resolution of 1 km (Hamann and Wang, 2018). We downloaded monthly climate variables from 2012 to 2015 and calculated an annual value for each variable at each plot based on the postfire year, October through September of the year following each fire. We proposed higher snow accumulation would insulate and protect seedlings during cold winter months, which would enhance recruitment success, while increased exposure to frost events could result in seedling mortality due to freezing. We expected recruitment to decline if climate moisture deficits were greater due to the moisture stress during the growing season.

We recorded seedlings and saplings (recruits) in burned and unburned plots. In burned plots, we recorded counts of all seedlings in the one-by-one-meter subplots, as well as seedlings in two of the four 5-by-5-meter quadrants. This nested strategy ensured a representative sample of recruits while maintaining sampling efficiency. For analysis, we aggregated seedling counts to a plot-level variable. In unburned (fire refugia) plots, we recorded all seedlings and saplings in the full 10-by-10-meter plot because of the low recruitment density. We defined saplings as a DBH range of 2.5–12.5 cm and a height less than canopy trees (less than 1.5 m). Seedlings had a DBH less than 2.5 cm with heights less than 15 cm. For each seedling/sapling in unburned plots, we recorded the number of branch whorls as a proxy for age (Kershaw et al., 2017). The wilderness objectives of the study area did not permit for destructive sampling, which through dendrochronological ring counting provides a more accurate method for estimating age in comparison to counting branch whorls or bud scars (Urza and Sibold, 2013; Harvey et al., 2016; Hankin et al., 2018).

2.3. Statistical analysis

We used multiple analyses to address our two research objectives, as outlined in Table 1. We tested the effect of each factor on seedling counts with an individual analysis (Table 1) due to the relatively small sample size. We used generalized linear models with a negative binomial distribution, using a log link, for all analyses. All models addressing our first objective included an offset variable that accounted for the area in which recruits were counted. To evaluate the effect of fire severity, we tested for a difference in mean number of recruits among our fire severity classes – crown, severe surface, light surface, and fire refugia. To determine the effect of the other contributing factors we regressed the number of recruits against each factor – outbreak severity, branch retention, open cones, cone abundance, soil exposure, frost events, snow accumulation, and climate moisture deficit and included an interaction term with fire severity.

For our second objective, we evaluated the potential drivers of recruitment in fire refugia. We tested for a difference in mean number of recruits across establishment periods – pre-outbreak (1990–1998),

Table 1
Study objectives, explanatory variables, and description of explanatory variables to evaluated drivers of seedling recruitment across a gradient of fire severity and in fire refugia. Due to our small sample size, we conducted multiple analyses to evaluate potential drivers of seedling recruitment. Sample sizes for each analysis are indicated in parentheses. Note that explanatory variable 1 × explanatory variable 2 indicates the inclusion of an interaction term, while explanatory variable 1 + explanatory variable 2 does not include an interaction term. All terms in the models were treated as fixed effects.

Explanatory variables	Description of explanatory variable
Objective 1: determine the drivers of seedling recruitment across gradient of fire severity	
Fire severity	Categorical variable including crown, severe surface, light surface, and unburned (n = 83)
Outbreak severity × fire severity	Outbreak severity is continuous proportion of prefire-killed trees (n = 63)
Branch retention × fire severity	Proportion of trees with remaining branch structure (n = 63)
Open cones × fire severity	Proportion of trees with open cones (n = 63)
Cone abundance × fire severity	Mean cones per tree/snag (n = 63)
Exposed soil × fire severity	Proportion of estimate exposed mineral soil for a plot (n = 63)
Snag fall × fire severity	Proportion of prefire snag fall (n = 63)
Frost events × fire severity	Total frost-free days one year after fire, calculated from October through September (n = 63)
Snow accumulation × fire severity	Total precipitation as snow one year after fire, calculated from October through September (n = 63)
CMD × fire severity	Total climate moisture deficit (CMD) one year after fire, calculated from October through September (n = 63)
Objective 2: determine the drivers of seedling establishment in fire refugia, areas only affect by MPB outbreaks	
Establishment periods	Categorical variable that divided seedlings into establishment periods based on estimated age, which included pre-outbreak (1990–1998), during-outbreak (1999–2006), and post-outbreak (2007–2015). (552 saplings across n = 20 plots)
Outbreak severity	Proportion of prefire-killed trees (n = 20)
Outbreak severity + branch retention	Proportion of trees with remaining branch structure (n = 20)
Outbreak severity + open cones	Proportion of trees with open cones (n = 20)
Outbreak severity + cone abundance	Mean cones per tree/snag (n = 20)

during outbreak (1999–2006), and post-outbreak (2007–2015) periods. We then regressed the number of recruits against each potential contributing factor – outbreak severity, branch retention, open cones, and cone abundance. All statistical analyses were conducted in R statistical computing software version 3.4.4 (R Development Team, 2018). We assessed model fit for all models through an inspection of the residuals and tested for overdispersion, which was not an issue in any of our models. Results were interpreted based on effect sizes and with p-values that ranged from 0.0 to 0.05 as evidence of a strong statistical effect and from 0.05 to 0.10 as suggestive of an effect. Data and code for analyses are available online (Talucci, 2019).

3. Results

Across the three fires sampled, lodgepole pine accounted for 96 percent of the canopy trees surveyed, with spruce accounting for the remaining 4%. We surveyed a total of 1191 trees, with 943 trees in 63 plots that experienced fire and 248 trees within 20 plots in fire refugia (Appx. Table S2). The mean prefire lodgepole pine mortality was 71% (range: 0–100% per plot). Fire caused mortality accounted for an additional 17% (range: 0–100%) for canopy lodgepole pine. Postfire seedling recruitment had a mean density of 12,202 recruits per hectare (range: 0–170,000), which exceeded the mean pre-disturbance canopy tree density of 1435 (range: 200–4200) trees per hectare (Appx. Table S2).

3.1. Drivers of recruitment across a gradient of fire severity

Across a gradient of fire severity, fire appeared to be a key mechanism and driver of lodgepole pine seedling recruitment. Plots that burned had significantly more recruits than fire refugia (Fig. 2). In plots that experienced crown fire and severe surface fire, seedling recruitment was over six times greater compared to fire refugia. In plots that experienced light surface fire, seedling recruitment was 2.94 times greater than plots in fire refugia (Fig. 2).

While accounting for fire severity, variability in seedling recruitment was primarily driven by cone abundance with other stand breakdown variables, soil, and climate variables having limited to no influence (Fig. 3, Appx. Fig. S2–S4). Seedling recruitment increased with increasing cone abundance (i.e. mean cones per tree) for plots that experienced severe and light surface fire. Seedling recruitment demonstrated an inverse relationship and declined with increasing cone

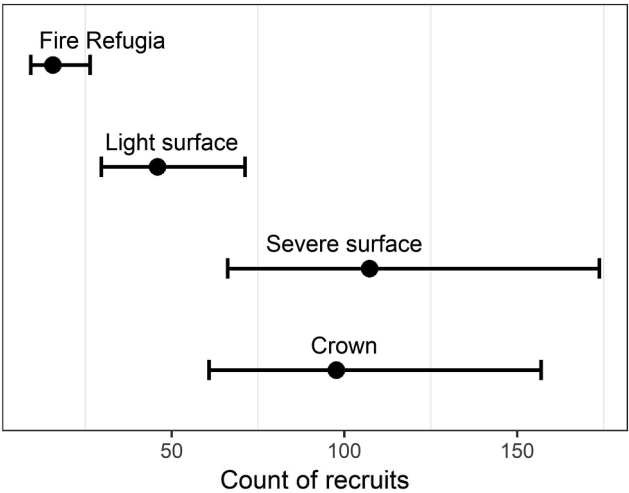


Fig. 2. Seedling recruitment across a gradient of fire severity. Estimated mean seedling counts per plot (100 m²) by fire severity classifications are shown with estimated 95% confidence intervals. Model estimates and response ratios are listed in Appx. Table S3.

abundance in plots that experienced crown fire. Seedling recruitment increased in response to the increasing proportion of trees with branches in plots that experienced light surface fire but showed no evidence of a relationship in plots that experienced severe surface and crown fire. All three postfire climate variables – frost events, snow accumulation, and climate moisture deficit, had three prominent outliers that influenced their statistical relationships and significance (Appx. Fig. S4). Frost events had no influence on seedling recruitment with and without outliers. Snow accumulation showed evidence of a direct negative relationship with seedling recruitment with the outliers included, but no evidence of a relationship when outliers were excluded. Climate moisture deficit showed no influence on seedling recruitment when outliers were included, but when outliers were excluded recruitment declined with increasing climate moisture deficit.

3.2. Drivers of recruitment across fire refugia

In fire refugia, recruits showed greater recruitment after the peak of the MPB outbreaks (Fig. 4). Recruits had a median age of 10 years

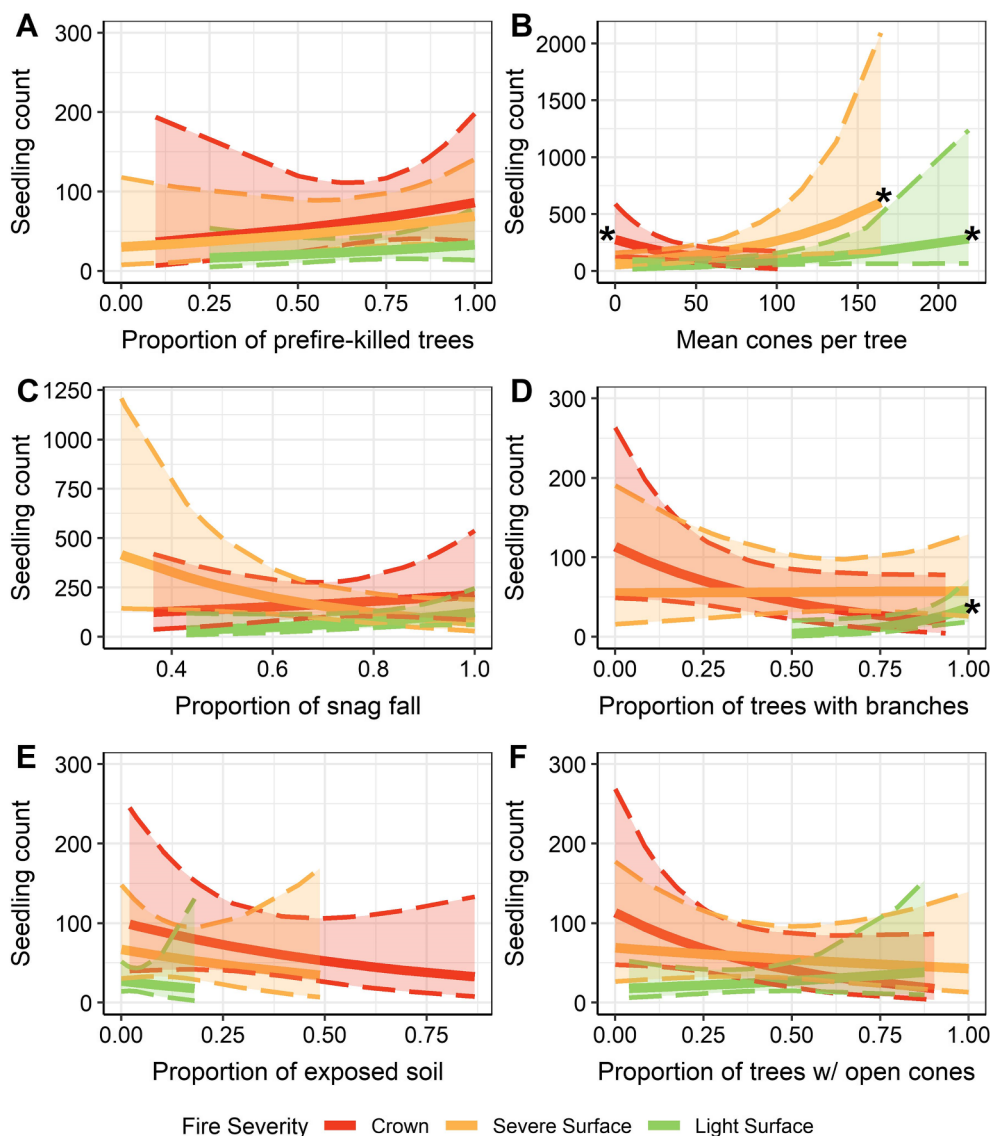


Fig. 3. Drivers of seedling recruitment across fire severity with statistical significance noted by an * and seedling counts per 100 m². The proportion of prefire killed trees exhibited no evidence of an effect on recruitment (A). Recruits declined with increasing mean cones per tree/snag in plots that experience crown fire and increased with increasing cones per tree in plots that experienced light surface fire (B). Snag fall exhibited no evidence of a relationship with recruitment, and plots that experienced severe surface fire, showed a declining relationship with increasing snag fall, although not statistically significant, the effect size is large (C). Only plots that experience light surface fire, showed evidence of an effect of proportion of trees with remaining branch structure on seedling recruitment (D). The proportion of exposed soil (E) and trees with open cones (F) show no evidence of an effect on recruitment. Model estimates are in Appx. Table S4 and with raw data plotted in Appx. Figs. S2–S3.

(range: 1–27), and a median diameter at their base of 1 cm (range: 1–10 cm). In the post-outbreak period, recruitment was just over three times greater compared to the pre-outbreak period (Fig. 4). Factors associated with stand breakdown – including outbreak severity, proportion of trees with open cones, proportion of trees with branches, and mean cones per tree (i.e. cone abundance), showed no statistical evidence of a relationship on the number of recruits (Fig. 5).

4. Discussion

Fire is a catalyst for lodgepole pine regeneration in these sub-boreal systems even after 10 years of stand breakdown and cone opening following MPB outbreaks. The pulse in postfire recruitment emphasizes the importance of wildfire as a mechanism for seed release and regeneration in these serotinous lodgepole pine forests. Fire severity and cone abundance were the primary drivers of recruitment density in burned stands, while substantial underlying tree mortality from the MPB outbreaks did not play a role. However, cone abundance may be acting as a partial proxy for MPB severity and stand breakdown variables. In fire refugia, a weak pulse in post-outbreak recruitment appeared to contribute to regeneration, but still had substantially lower densities of seedlings and saplings in comparison to their burned counterparts. Patches of the landscape that experience lower

recruitment rates may lead to delayed recovery of lodgepole pine that could contribute important structural or demographic heterogeneity to the system (Seidl et al., 2016), but also offer possible foot-holds for slow shifts in composition and structure in response to global change (Stralberg et al., 2018).

4.1. Drivers of recruitment across a gradient of fire severity

Wildfire is a key mechanism for seedling establishment, with burned plots experiencing higher recruitment rates than fire refugia. Seedling density was highest in plots that experienced severe surface fire, which aligns with previous research that examined beetle-fire interactions in subalpine forests (Harvey et al., 2014a; 2014b). Areas burned by severe surface and crown fires exhibited similar recruitment rates, potentially because they produced similar conditions for seed release and site conditions conducive for germination. While the canopy seedbank is reduced through relocation to the forest floor from stand breakdown (Teste et al., 2011), lodgepole pine seems to generally produce enough cones and seeds that regeneration is not hindered even when disturbances recur within a short interval.

Stand breakdown can influence the structural and ecological legacies that persist through short-interval MPB-fire disturbances that act as controls for regeneration and contribute to postfire landscape

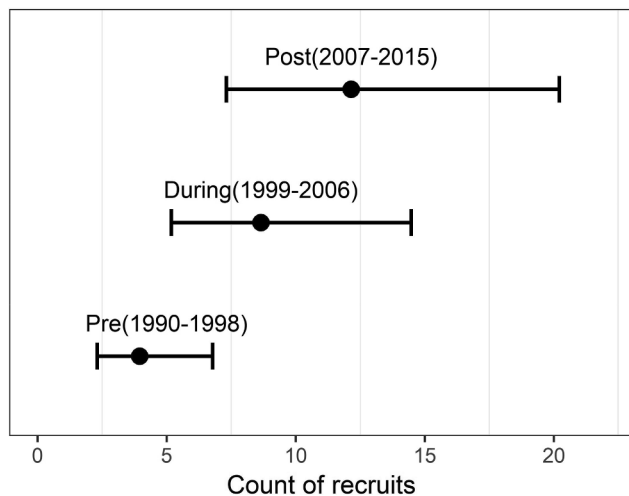


Fig. 4. An examination of recruits during establishment periods: pre-outbreak (1990–1998; nine year interval), during outbreak (1999–2006; eight years), and post-outbreak (2007–2015; nine year) in fire refugia. Estimated mean count of seedlings per plot (100 m²) that established pre-, during-, or post-outbreak are shown with estimated 95% confidence intervals. Model estimates and response ratios are in Appx. Table S5.

heterogeneity. The proportion of prefire mortality (dominantly from MPB), remaining branches, cone abundance, open cones, and snag fall were factors that we hypothesized might influence seedling recruitment postfire and could potentially be correlated. Cone abundance was the only clear driver of recruitment amongst these factors and was not correlated with MPB severity, but it was moderately correlated with remaining canopy structure ($r = 0.61$). In plots that experienced severe and light surface fire, recruitment had a direct positive relationship with cone abundance. In plots that experienced crown fire, recruitment declined with increasing cone abundance, which may point to instances where overheating kills seeds (Alexander and Cruz, 2012) or other correlating effects that we were unable to tease apart. These findings highlight that serotiny is a key trait for lodgepole pine regeneration even after stand breakdown from MPB outbreaks (Schoennagel et al., 2012; Harvey et al., 2014a, 2014b) and that lodgepole pine seedling density varies with fire severity (Turner et al., 1997).

Exposed mineral soil did not systematically influence seedling recruitment in our burned landscapes, even though it is known to be important for lodgepole pine germination and recruitment (Lotan et al., 1985). Increased exposure of mineral soil is often associated with significant canopy mortality, which frees up available resources for seedling regeneration including water, nutrients, and full sun (Lotan et al., 1985). The exposed mineral soil variable was based on a visual estimate of percent coverage and was measured in early successional

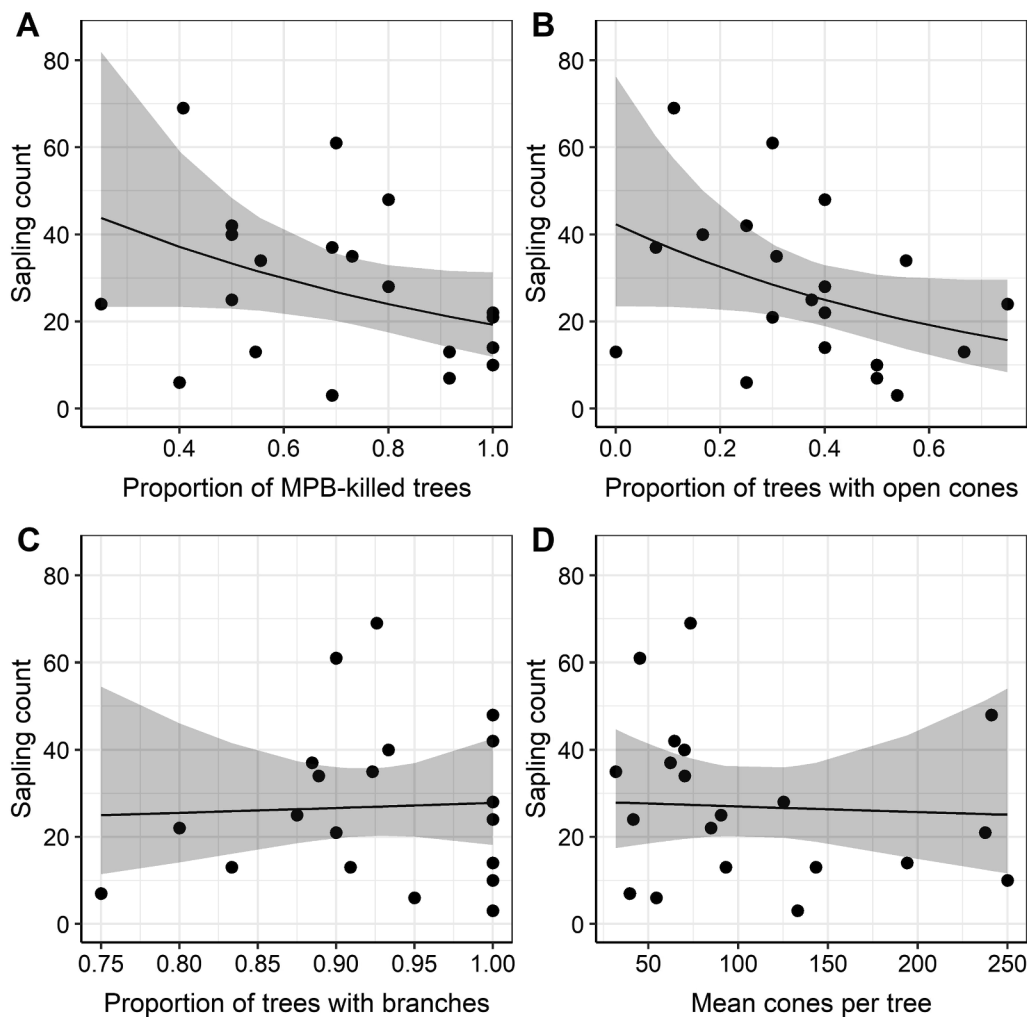


Fig. 5. Recruitment in unburned plots (fire refugia) as a function of stand breakdown including outbreak severity, remaining branches, open cones, and mean cones per tree/snag in fire refugia. There was no evidence of a statistical effect on recruitment from any of the stand breakdown variables. Recruitment declined with increasing outbreak severity (A). Recruitment declined with increasing cone opening (B). Recruitment was unrelated to the proportion of trees with branches (C). Recruitment was unrelated to the average number of cones per stem (D). Model estimates are in Appx. Table S6.

forest communities two to four years postfire. The true values of this could have been obscured or under-estimated due to vegetative regrowth. We also did not specifically measure microsite characteristics such as soil type, moisture levels, or nutrient levels, which can also influence seedling success and/or failure (Batllori et al., 2009; Walck et al., 2011) and should be considered in future research.

Postfire climate conditions are relevant drivers of seedling recruitment and success. Frost events, snow accumulation, and water stress are known to influence seedling success in other subalpine systems (Inouye, 2000; Batllori et al., 2009; Walck et al., 2011; Harvey et al., 2016; Renard et al., 2016; Clark et al., 2017; Hansen et al., 2018). Due to the life zone equivalence of subalpine and sub-boreal forests (Holdridge, 1947), we expected similar climate controls on seedling response to apply. In the year following each fire, there were fewer frost events, less snow, and a greater climate moisture deficit compared to the climate normals data for 1961–2000 and 1981–2010 in ways that indicated anomalously warm and dry conditions (Appx. Fig. S5). However, the narrow geographical range of our field sites and limited temporal comparison (e.g., three years for contrast), combined with the coarse resolution of the climate data provided a low range of variability within our data and limited our ability to discern effects. Snow accumulation and climate moisture deficit both indicated some control over seedling recruitment, but this varied with the inclusion or exclusion of prominent outliers. With outliers removed, seedling recruitment declined with increasing climate moisture deficit. This general trend aligns with previous findings that identified drought effects as constraints on postfire recruitment (Harvey et al., 2016; Petrie et al., 2016). Under future conditions, sub-boreal landscapes may begin to experience increased stress as the climate warms, which could limit regeneration (Dale et al., 2001; Flannigan et al., 2009; Gonzalez et al., 2010). The combination of disturbances that recur at short intervals with sequential drought events may open opportunities for range shifts in species distributions and forest composition (Johnstone et al., 2016; Stralberg et al., 2018).

4.2. Drivers of recruitment across fire refugia

In fire refugia, we observed greater recruitment during- and post-outbreak compared to the pre-outbreak period. In the post-outbreak (2007–2015) period, recruitment establishment was three times greater than before the outbreaks (1990–1998). Data collection solely captured recruits that existed at time of sampling and does not account for potential mortality over time, which could account for this discrepancy between establishment periods. Further, counts of branch whorls as a proxy for age are not as accurate as destructive sampling and tend to under estimate ages (Urza and Sibold, 2013; Kershaw et al., 2017; Hankin et al., 2018), which could mean that there are actually more recruits during and post-outbreak than we are currently accounting for. In central interior BC, lodgepole pine is known to have high levels of serotiny (Carlson, 2008), and although non-serotinous lodgepole pine exists, it is often associated with lower density, mixed-aged stands (DeLong and Kessler, 2000). The timing of recruitment suggests that stands in fire refugia were likely producing serotinous cones that senesced and opened as a function of outbreak mortality, based on Teste et al. (2011) analysis of seed loss and/or xeriscent-cueing. However, the population could also include a chronic level of non-serotinous individuals; our postfire data collection did not specifically aim to quantify levels of serotiny.

Gradients in factors associated with stand breakdown, e.g. MPB mortality, branch structure, cone opening, and cone abundance, did not show evidence of an effect on seedling recruitment among our fire refugia plots. In the shade of snags and with no exposed mineral soil, recruits appeared relatively stunted and less robust compared to their postfire counterparts, likely due to resource competition and lower light levels as described by Astrup et al. (2008) and consistent with findings by de Ville (2013) in southern interior BC. In this post-MPB landscape,

fire refugia augment landscape heterogeneity for the broader mosaic by supporting forest patches with lower density and a mixed-age structure of lodgepole pine.

4.3. Management considerations

Our findings contribute insight on forest management for both protected areas (e.g., provincial parks, national parks) and lands managed for timber production (public/crown and private lands) that have been impacted by recent MPB outbreaks. Fire is a critical mechanism for seed release and recruitment for lodgepole pine dominated forests that experience severe MPB outbreaks. Fire serves to jump-start and accelerate the regeneration of lodgepole pine dominated landscapes with substantial mortality from outbreaks, which is prudent if/when managing for forest regeneration. Within burn mosaics dominated by high severity fire, the arrangement and size of fire refugia are an important consideration for land management and conservation, because they augment landscape heterogeneity. In the landscapes studied here, fire refugia encompassed gradients of composition, structure, and succession including post-outbreak early seral forests and mature forests dominated by live trees. These patches of fire refugia can act as micro ecosystems that provide structural and ecological diversity (Meddens et al., 2018) including seed sources (Turner et al., 1997) and habitat for wildlife (Robinson et al., 2013). The unevenness in recruitment across gradients of wildfire and MPB outbreak severity, including fire refugia, contribute to landscape heterogeneity that may mediate future outbreaks and enhance forest resilience.

5. Conclusions

Sub-boreal lodgepole pine forests are generally resilient when MPB outbreaks and wildfire recur in short intervals, with wildfire serving as a key mechanism for seed release and forest regeneration. The postfire burn mosaic contributes to landscape heterogeneity through variability in stand density and age classes. In BC, the recent fire seasons, 2017 and 2018, have been notable in social costs and wildfire footprints with a record number of evacuees and a historic number of hectares burned. The 2017 and 2018 fire seasons have also burned large portions of central interior BC that include forests impacted by MPB outbreaks. Our results point to the resilience of these forests to short-interval disturbances, however as these forests are regenerating, they are potentially vulnerable to future increases in fire activity that could lead to young forests experiencing a reburn event, where fires recur in short intervals. Early seral lodgepole pine forests on these landscapes often will not produce serotinous cones for another couple of decades limiting the seedbank development (Lotan and Perry, 1983) and constraining their regeneration capacity (Johnstone et al., 2016). Further evaluation and monitoring of forest regeneration in relation to future disturbance dynamics that may influence forest resilience will enhance the ability to characterize successional trajectories for these landscapes under short-interval disturbance regimes and global change.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2019.117500>.

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